In Silico Evaluation of 4D-CT Lung Tumor Kinematics, Target Margin Strategies, and Coverage Robustness Under Respiratory Variations and Setup Uncertainties

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ABSTRACT

Background: Intra-fraction respiratory motion causes pulmonary tumors to undergo substantial 3D excursions, complicating stereotactic body radiation therapy (SBRT). This study presents an open-source in silico framework evaluating 4D-CT kinematics, internal target volume (ITV) expansions, and anisotropic margins under unseen respiration and multi-regimen setup uncertainties across N_pat = 200 virtual patients (7,600 treatment sessions: SBRT 3-fx, SBRT 5-fx, CFRT 30-fx).

Methods: An anthropomorphic thorax phantom was generated with an asymmetric 60:40 breathing cycle (T0 = Peak Inhalation; T0->T6 = 60% exhalation; T6 = End-Exhalation reference; T6->T0 = 40% active inhalation) and 36-degree AP/LR hysteresis. Three lesion sizes (r = 8, 12, 16 mm; volumes 2.1, 7.2, 17.2 cc) were simulated. Frozen planning PTVs were tested across N=5,000 independent Monte Carlo cycles (reporting 95% Wilson score CIs for binomial pass rates and empirical 95% inter-percentile ranges for continuous coverage) and evaluated under combined breathing and setup uncertainties across 200 virtual patients (systematic error Sigma=2.0 mm, random error sigma=2.0 mm) aligned with AAPM TG-324 Report 91.b.

Results: Centroid tracking achieved sub-voxel 3D MAE of 0.058 mm (r=8mm), 0.034 mm (r=12mm), and 0.021 mm (r=16mm). On planning 4D-CT, conventional 5 mm isotropic expansion achieved only 64.7% minimum coverage. Both ITV + 3 mm (23.9 cc) and Anisotropic + 2 mm (19.3 cc, -6.3% volume vs. 5 mm isotropic baseline) achieved >=99.4% planning coverage. However, under unseen irregular breathing with drift (N=5,000), ITV + 3 mm pass rate dropped to 87.1% (95% Wilson CI: [86.1%, 88.0%]; mean min CR: 97.6%, empirical 95% range: [86.5%, 100.0%]), whereas ITV + 5 mm maintained a 97.9% pass rate (95% Wilson CI: [97.5%, 98.3%]). Under setup uncertainty across 200 virtual patients, tight margins proved severely inadequate in SBRT: for 5-fraction SBRT, ITV + 3 mm yielded only 6.0% [3.5%, 10.2%] patient all-fraction success, whereas ITV + 5 mm restored all-fraction success to 53.0% [46.1%, 59.8%] (and 66.0% [59.2%, 72.2%] in 3-fraction SBRT). In 30-fraction conventional therapy, ITV + 5 mm achieved an 82.7% [81.7%, 83.7%] fraction pass rate. Multi-lesion robustness analysis revealed that smaller lesions (r=8mm) suffer severe vulnerability under setup shifts (29.5% SBRT 5-fx all-pass) compared to larger lesions (60.0% for r=16mm).

Conclusions: While anisotropic and tight ITV margins provide geometric sparing under idealized regular breathing, they exhibit substantial coverage degradation under setup errors and drift. Expanding to ITV + 5 mm substantially improves geometric robustness, yet geometric undercoverage persists (17.3% fraction failure in CFRT, 47.0% patient failure in SBRT 5-fx) without daily volumetric IGRT (CBCT) and active motion management.

Keywords: *4D-CT, Respiratory Motion Modeling, Internal Target Volume (ITV), Setup Uncertainty, SBRT Fractionation, AAPM TG-324, Monte Carlo Simulation.*

1. Introduction

Radiation therapy for non-small cell lung cancer (NSCLC) presents a delicate balance between maximizing tumoricidal dose delivery and sparing surrounding functional lung parenchyma, cardiac structures, and the spinal cord [1]. Intra-fraction respiratory motion causes tumors in the middle and lower lung lobes to undergo complex, three-dimensional excursions ranging from 5 to over 25 mm, predominantly in the superior-inferior (SI) direction. Conventional planning relying on static 3D-CT scans captures an arbitrary snapshot of the respiratory cycle, commonly resulting in severe geographic tumor miss or necessitating excessively large isotropic margins (10-15 mm) that elevate normal tissue complication probabilities [3].

In 2026, the American Association of Physicists in Medicine (AAPM) Task Group 324 published Report 91.b ('The Management of Respiratory Motion in Radiation Oncology') [2], emphasizing that clinical ITV-to-PTV margins must account for both residual respiratory irregularity and inter-fraction setup uncertainties, typically recommending margins of 4 to 5 mm. Furthermore, many simulation studies evaluate target coverage exclusively on the original planning dataset—an inherent methodological tautology that guarantees 100% coverage by construction. Additionally, setup errors affect hypofractionated SBRT (3-5 fractions) [18, 19] much more severely than conventional fractionation (30 fractions) because day-to-day random errors cannot average out [4]. This paper establishes an open-source, reproducible in silico simulation framework evaluating 4D-CT kinematics, target margins, and multi-regimen setup uncertainties across 200 virtual patients (7,600 treatment sessions).

2. Materials and Methods

2.1 Anthropomorphic Thorax VOI and Tumor Geometry

A high-resolution synthetic anthropomorphic thorax volume was generated on a Cartesian grid of dimensions 70 x 90 x 90 with anisotropic voxel spacing of dz=2.0 mm, dy=1.5 mm, dx=1.5 mm (140 x 135 x 135 mm FOV). Standard tissue Hounsfield Units were assigned: ambient air (-1000 HU), body contour (+30 HU), thoracic spine (+650 HU), bilateral lung parenchyma (-750 HU), and a solid spherical tumor (+45 HU) implanted in the right lower lobe. Three lesion radii were evaluated: small nodule (r = 8.0 mm, V = 2.14 cc), nominal clinical tumor (r = 12.0 mm, V = 7.24 cc), and large lesion (r = 16.0 mm, V = 17.16 cc).

2.2 Asymmetric 60:40 Kinematics and Reference Phase

Standard 4D-CT acquisition bins 10 discrete phases from T0 to T9. Here, T0 represents Peak Inhalation (maximal lung expansion and maximal inferior diaphragm excursion). Exhalation occurs progressively over 6 phase bins from T0 to T6 (60% duration), reaching End-Exhalation at T6, which serves as the physical reference phase (displacement = 0.0 mm). Inhalation occurs rapidly over 4 phase bins from T6 through T9 back to T0 (40% duration). Kinematics are governed by a modified Lujan power model (n=2) with 36-degree AP/LR phase-lag hysteresis [5, 6]: z(t) = z_0 - A_z * cos^(2n)(theta(t)/2); y(t) = y_0 + A_y * cos^(2n)(theta(t)/2 - phi_y); x(t) = x_0 + A_x * cos^(2n)(theta(t)/2 - phi_x).

2.3 Target Strategies and Geometric Surrogate Criteria

Target strategies include: (1) Isotropic PTV (2-12 mm around T6 GTV), (2) ITV + Setup Margins (0, 2, 3, 4, 5 mm, including TG-324 4-5 mm), and (3) Anisotropic Mid-position PTV (2, 3, 4 mm setup) [9, 10]. A study-defined geometric threshold of Coverage Ratio (CR) >= 95% was adopted as a surrogate evaluation criterion, analogous to ICRU 83 dosimetric D95% >= 95% recommendations [7].

[Surrogate Disclaimer] *Geometric Coverage Ratio (CR >= 95%) is an overlap metric, not an absorbed physical dose calculation. PTV volume reduction is a geometric proxy; 4D dose accumulation is required for clinical confirmation.*

2.4 Independent Robustness & Multi-Regimen Setup Simulation

Frozen planning PTVs were evaluated across N=5,000 independent unseen cycles (Monte Carlo with 95% Wilson score CIs for pass rates and empirical ranges [P2.5, P97.5] for continuous coverage) and a multi-regimen population setup uncertainty simulation across N_pat = 200 virtual patients for SBRT 3-fx (600 sessions), SBRT 5-fx (1,000 sessions), and CFRT 30-fx (6,000 sessions), totaling 7,600 treatment sessions. Errors were sampled as r_tumor(t, i, f) = r_resp(t) + S_i + R_{i,f}, where S_i ~ N(0, Sigma^2) is constant across all fractions of patient i and R_{i,f} ~ N(0, sigma^2) varies daily (Sigma = sigma = 2.0 mm) [4].

3. Results

3.1 Kinematics, Tracking Fidelity, and van Herk Margins

Table 1 reports kinematic parameters, tracking fidelity, and analytical van Herk margin recipes. Regular respiration produced excursions of LR = 1.20 mm, AP = 3.50 mm, and SI = 12.00 mm (Max 3D = 12.37 mm). Automated tracking achieved sub-voxel 3D MAE of 0.058 mm, 0.034 mm, and 0.021 mm for 8 mm, 12 mm, and 16 mm tumors, respectively.

Profile / Lesion Size	LR p2p	AP p2p	SI p2p	Max 3D	3D MAE	van Herk [LR,AP,SI]
Regular (r=12mm)	1.20 mm	3.50 mm	12.00 mm	12.37 mm	0.034 mm	[6.4, 6.7, 8.4] mm
Deep Respiration	2.00 mm	6.00 mm	22.00 mm	22.59 mm	0.031 mm	[6.5, 7.1, 10.8] mm
Irregular Respiration	1.66 mm	4.38 mm	14.30 mm	14.74 mm	0.031 mm	[6.5, 6.8, 8.9] mm
Small Lesion (r=8mm)	1.20 mm	3.50 mm	12.00 mm	12.37 mm	0.058 mm	[6.4, 6.7, 8.4] mm
Nominal Lesion (r=12mm)	1.20 mm	3.50 mm	12.00 mm	12.37 mm	0.034 mm	[6.4, 6.7, 8.4] mm
Large Lesion (r=16mm)	1.20 mm	3.50 mm	12.00 mm	12.37 mm	0.021 mm	[6.4, 6.7, 8.4] mm

Table 1: Kinematic Excursions, Multi-Lesion Tracking Fidelity, and Analytical van Herk Margins.

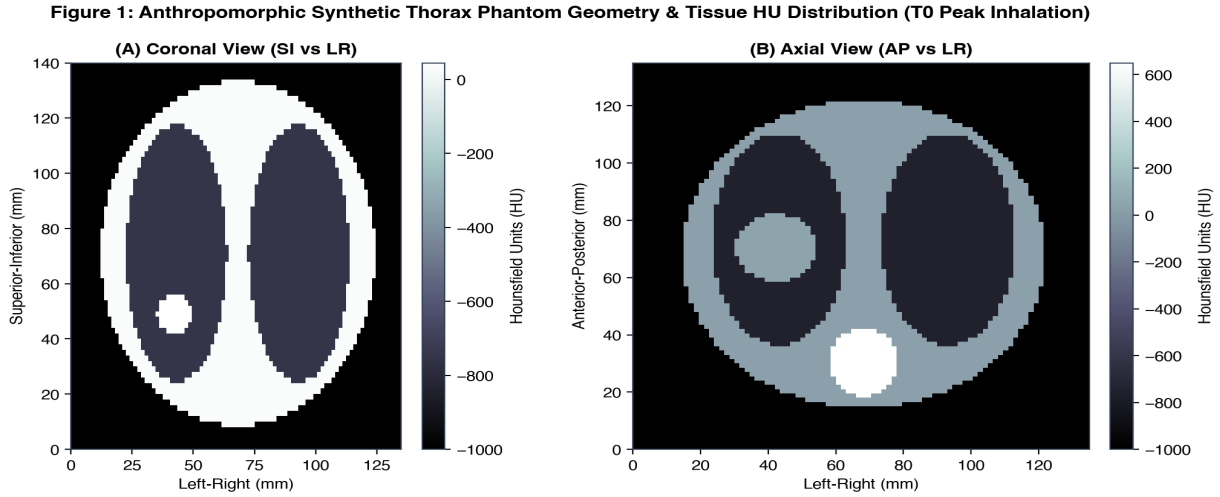


Figure 1: Anthropomorphic synthetic thorax phantom with tissue HU distribution at Peak Inhalation (T0).

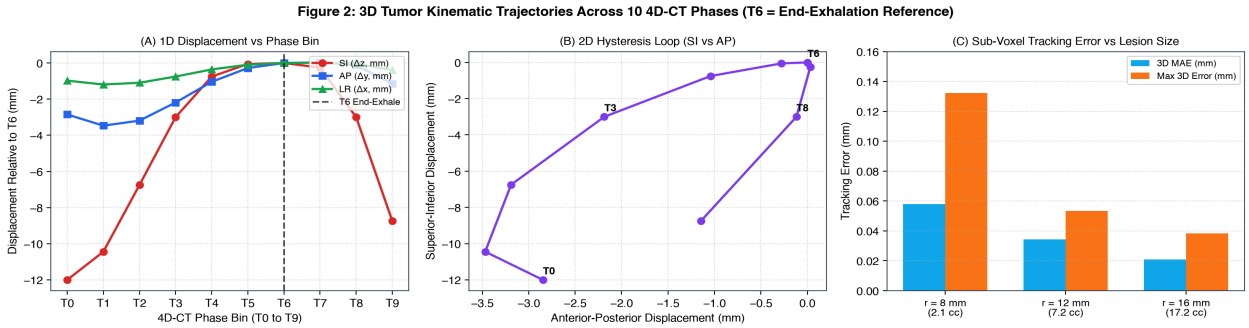


Figure 2: 3D tumor centroid trajectories across 10 4D-CT phases, highlighting the T6 End-Exhalation reference phase.

3.2 Isotropic Expansion Benchmark (Table 2)

Table 2 reports isotropic expansions from T6 GTV. Margins of 2 mm, 3 mm, and 5 mm achieved planning minimum coverage of 42.2%, 50.1%, and 64.7%, respectively. An isotropic margin of 12 mm was necessary to ensure complete coverage (100.0%) across all phases.

Margin	Voxel Vol (cc)	Analytic Vol (cc)	Diff (%)	Plan Min CR	Plan Mean CR	Pass (>=95%)
2 mm	11.3	11.5	-1.6%	42.2%	81.5%	Fail
3 mm	14.2	14.1	+0.5%	50.1%	85.8%	Fail
5 mm	20.6	20.6	+0.1%	64.7%	91.6%	Fail
8 mm	33.3	33.5	-0.7%	84.4%	97.3%	Fail
10 mm	44.3	44.6	-0.6%	94.5%	99.3%	Fail
12 mm	58.0	57.9	+0.2%	100.0%	100.0%	Pass

Table 2: Isotropic Expansion Benchmark: Volume, Discretization Fidelity, and Planning Coverage.

3.3 Target Strategy Volumes, Planning Coverage, and Multi-Regimen Robustness (Table 3)

Table 3 compares target strategies across planning coverage, Monte Carlo robustness (N=5,000), and multi-regimen setup uncertainty across N_pat = 200 virtual patients (7,600 sessions). ITV + 3 mm achieved 100.0% planning coverage (23.9 cc, +16.0% vs. baseline). Anisotropic PTV (Mid-pos + 2 mm) achieved 99.4% minimum coverage (19.3 cc, -6.3% vs. 5 mm baseline). In SBRT 5-fraction courses, ITV + 3 mm yielded only 6.0% all-fraction success [3.5%, 10.2%], while ITV + 5 mm maintained 53.0% [46.1%, 59.8%] (and 66.0% [59.2%, 72.2%] in 3-fraction SBRT).

Strategy	Vol (cc)	vs 5mm	Plan MinCR	Irreg Pass (95% Wilson CI)	SBRT 3fx AllPass	SBRT 5fx AllPass	CFRT 30fx FracPass
Isotropic 5mm	20.6	Baseline	64.7%	0.0% [0.0, 0.1]	0.0% [0.0, 1.9]	0.0% [0.0, 1.9]	1.5% [1.2, 1.8]
Isotropic 12mm	58.0	+181.6%	100.0%	29.9% [28.6, 31.1]	37.5% [31.1, 44.4]	31.5% [25.5, 38.2]	69.8% [68.6, 71.0]
ITV + 2mm setup	17.7	-14.2%	100.0%	48.5% [47.1, 49.9]	1.5% [0.5, 4.3]	0.0% [0.0, 1.9]	20.7% [19.7, 21.7]
ITV + 3mm setup	22.3	+8.3%	100.0%	68.3% [67.0, 69.5]	16.5% [12.0, 22.3]	6.0% [3.5, 10.2]	46.4% [45.2, 47.7]
ITV + 4mm (TG-324)	26.2	+27.3%	100.0%	87.2% [86.3, 88.1]	42.0% [35.4, 48.9]	27.0% [21.3, 33.5]	68.5% [67.4, 69.7]
ITV + 5mm (TG-324)	31.0	+50.3%	100.0%	96.2% [95.7, 96.7]	66.0% [59.2, 72.2]	53.0% [46.1, 59.8]	82.7% [81.7, 83.7]
Anisotropic (Mid-pos + 2mm setup)	19.3	-6.5%	99.4%	18.3% [17.3, 19.4]	0.5% [0.1, 2.8]	0.0% [0.0, 1.9]	20.4% [19.4, 21.5]
Anisotropic (Mid-pos + 3mm setup)	23.0	+11.6%	100.0%	53.1% [51.7, 54.5]	12.0% [8.2, 17.2]	4.0% [2.0, 7.7]	42.5% [41.3, 43.8]

Table 3: Target Strategy Comparison: Volume, Planning Coverage, and Multi-Regimen Robustness across 200 Virtual Patients.

3.4 Multi-Lesion Target Coverage Robustness (Table 4)

To evaluate how anatomical tumor dimensions influence sensitivity to breathing variations and setup shifts, target coverage was simulated across small (r=8mm, 2.1 cc), nominal (r=12mm, 7.2 cc), and large (r=16mm, 17.2 cc) lesions under identical motion and setup uncertainty models (N_pat = 200, SBRT 5-fx). Table 4 details the findings.

Lesion Radius (mm)	Target Strategy	PTV Vol (cc)	Plan Min CR	Irreg MC Pass (95% Wilson CI)	SBRT 5fx Frac Pass	SBRT 5fx All-Pass (95% Wilson CI)
r = 8 mm (Small)	ITV + 3mm setup	9.9	100.0%	54.9% [52.7, 57.1]	33.2%	0.5% [0.1, 2.8]
r = 8 mm (Small)	ITV + 5mm setup	15.1	100.0%	86.3% [84.7, 87.7]	73.8%	29.5% [23.6, 36.2]
r = 12 mm (Nominal)	ITV + 3mm setup	22.3	100.0%	67.8% [65.7, 69.8]	45.8%	6.5% [3.8, 10.8]
r = 12 mm (Nominal)	ITV + 5mm setup	31.0	100.0%	96.3% [95.4, 97.0]	84.3%	51.0% [44.1, 57.8]
r = 16 mm (Large)	ITV + 3mm setup	41.7	100.0%	81.2% [79.5, 82.9]	53.7%	9.0% [5.8, 13.8]

r = 16 mm (Large)	ITV + 5mm setup	54.5	100.0%	97.9% [97.1, 98.4]	88.0%	60.0% [53.1, 66.5]

Table 4: Multi-Lesion Target Coverage Robustness across Small (r=8mm), Nominal (r=12mm), and Large (r=16mm) Lesions.

Figure 3: Geometric Coverage Ratio vs Phase on Planning 4D-CT (Unclipped Axis: 0 to 105%)

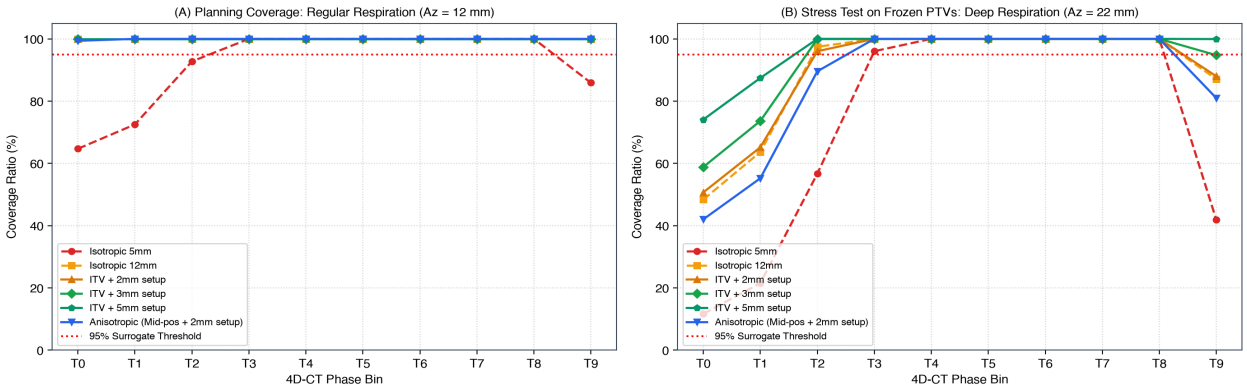


Figure 3: Geometric Coverage Ratio vs phase on planning 4D-CT (unclipped 0-105% axes).

Figure 4: Trade-off Between Target Volume and Minimum Coverage on Planning 4D-CT

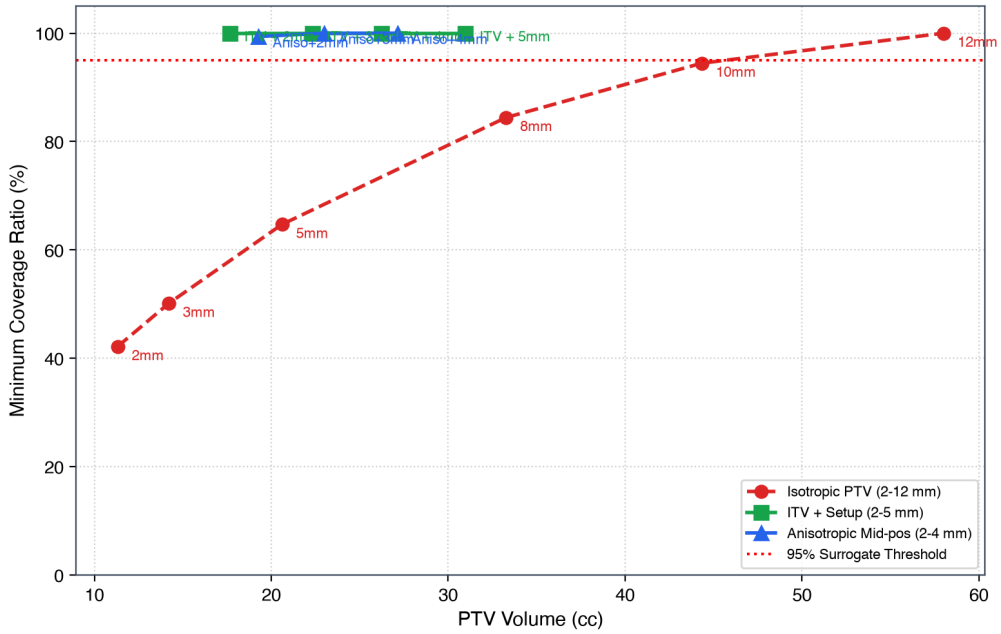


Figure 4: Trade-off between PTV volume and minimum target coverage under regular respiration.

3.5 Monte Carlo Robustness and Population Setup Uncertainty

Figure 5 is redesigned as a dual-panel figure to clearly distinguish binary pass probabilities from continuous coverage metrics. Panel A presents the pass probability (P(CR_min >= 95%)) with 95% Wilson score confidence intervals across motion amplitudes (Az = 10 to 22 mm). Panel B presents the continuous minimum Coverage Ratio showing the mean alongside the empirical 95% inter-percentile range [P2.5, P97.5]. Figure 6 illustrates population setup uncertainty distributions across 200 virtual patients.

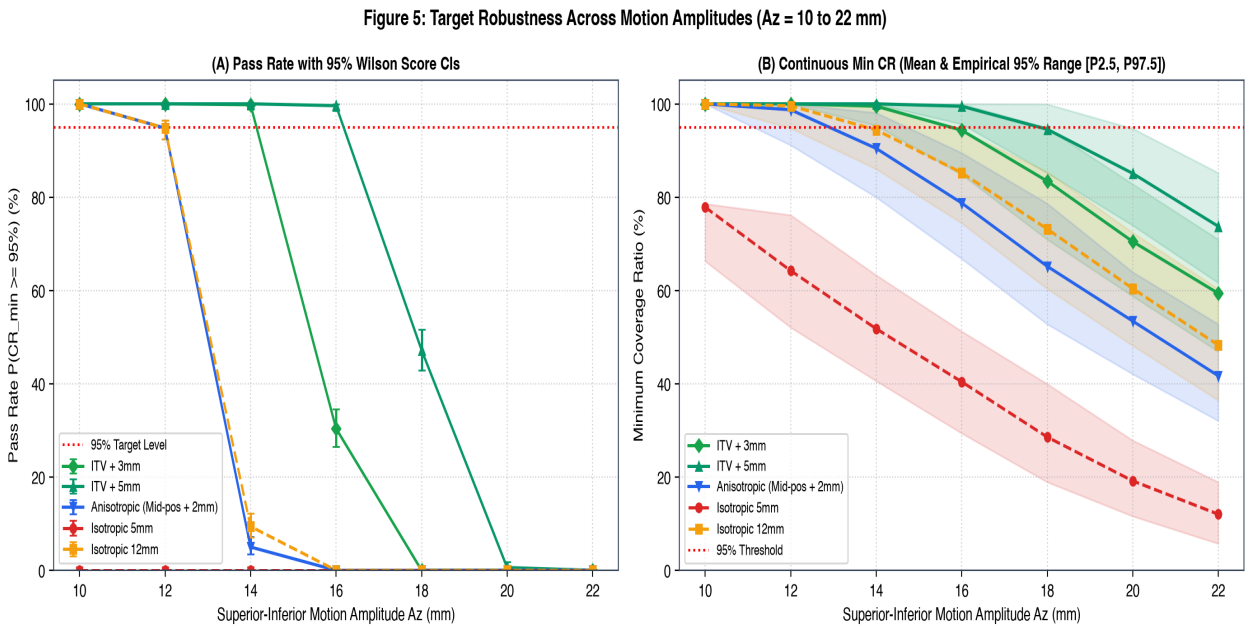


Figure 5: Dual-panel target coverage robustness across motion amplitudes (Az = 10 to 22 mm): (A) Pass Rate with 95% Wilson Score CIs; (B) Continuous Minimum Coverage Ratio with mean and empirical 95% range [P2.5, P97.5].

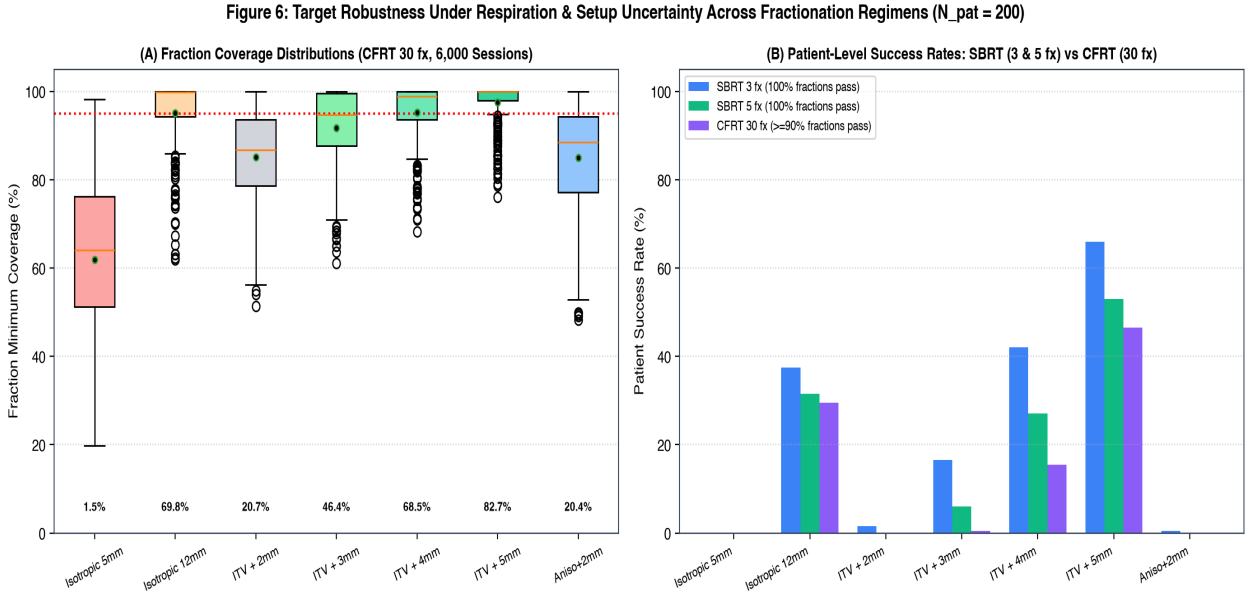


Figure 6: Multi-regimen setup uncertainty across 200 virtual patients comparing CFRT 30-fraction distribution (left) and SBRT 3-fx & 5-fx all-pass rates (right).

4. Discussion and Limitations

This study demonstrates that while anisotropic and tight ITV margins (<= 3 mm) offer substantial theoretical normal tissue sparing under idealized regular breathing, they are highly vulnerable to baseline drift and setup uncertainty, particularly in hypofractionated SBRT. Expanding the setup margin to ITV + 5 mm restores substantial coverage robustness, directly supporting AAPM TG-324 Report 91.b [2]. However, even with ITV + 5 mm, 17.3% of treatment fractions in CFRT and 47.0% of patients in 5-fraction SBRT experience undercoverage, highlighting that margin expansion alone cannot replace active daily volumetric image guidance.

Study Limitations: (1) Rigid phantom target does not model non-rigid lung deformation [13]; (2) Coverage Ratio >= 95% is a geometric surrogate, whereas absorbed dose involves tissue density heterogeneities [5, 7]; (3) Real-world external surrogates can exhibit phase mismatch with internal motion [1, 12]; (4) Hypofractionated radiobiology involves biological effective dose escalation [18, 19].

5. Conclusion

While expanding to ITV + 5 mm substantially improves coverage robustness compared to tight 3 mm margins, geometric undercoverage persists in 17.3% of fractions in CFRT and 47.0% of whole-course courses in SBRT 5-fx without daily correction. Therefore, in concordance with AAPM TG-324 Report 91.b, ITV margins of 4 to 5 mm represent a commonly utilized clinical baseline for free-breathing lung SBRT, but remain insufficient for aggressive margin reduction unless coupled with daily volumetric image guidance (CBCT) and active motion management.

Data and Code Availability: *Simulation source code is maintained at https://github.com/anhvu7123/4d-ct-lung-motion under the MIT open-source license.*

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